

Mohamed Tlili

GIS Engineer | Cartography Specialist

Tunisian | (+216) 96 806 719 | tlilixmed@gmail.com | tliligis.me | linkedin.com/in/tlilixmed

Geomatics engineer specializing in GIS, cartography, and LiDAR processing, with expertise in geodatabase design, python automation, and spatial data production for mineral exploration.



Professional Experience

WOLO Engineering | Ariana, Tunisia | Dec 2023 - Present

Team Lead

Dec 2025 - Present

- Led a team designing and deploying a cartographic geodatabase in ArcGIS Pro with custom data dictionaries; built automated data-validation pipelines (Python + geodatabase rules) and integrated restitution deliverables for specification compliance.
- Coordinated project delivery and deadline adherence, ensuring team cohesion and continuous alignment with client requirements.

Cartographic Study Specialist

Dec 2023 - Dec 2025

- Processed and classified LiDAR point clouds (ground, vegetation, buildings) using MicroStation and TerraScan; generated digital terrain models DTM/DSM and performed noise filtering.
- Conducted GCP placement and dense point-cloud processing in Agisoft Metashape; generated DTM/DSM and produced high-resolution orthomosaics.
- Updated fiber-optic and infrastructure data in Smallworld GE; coordinated with cross-functional teams and clients to align spatial data with AutoCAD plans.
- Conducted precise digitization and feature extraction from drone imagery, ensuring the accuracy and reliability of cartographic deliverables.

GIS Consultant Freelance - Mineral Exploration

Nov 2025 - Present

- Produced thematic maps, cartographic layouts, and technical documents for land analysis, mineral tenure, and geological exploration projects.
- Identified staking opportunities and detected claim overlaps by cross-referencing geological, cadastral, and tenure datasets (PLSS, BLM, Crown concessions, LTSA, mining claims) for targeted mineral exploration.
- Converted GIS data to CAD and created professional AutoCAD plans with complete cartographic styling, managing coordinate transformations and datum conversions in ArcGIS Pro.
- Automated repetitive geoprocessing and file management tasks using Python; built custom ArcGIS toolboxes to streamline spatial data workflows.

Skills

GIS & Cartography: ArcGIS Pro, QGIS (advanced), Smallworld GIS, Agisoft Metashape | **CAD:** AutoCAD, MicroStation, FME | **Programming:** Python (geoprocessing, automation, toolboxes), LAS tools

Projects

- LAS File Manager:** Developed a Python application to split and merge LAS files by X/Y axes or grid, with interactive HTML map preview and real georeferencing; automated workflow using LAsTools for significant time savings in LiDAR data preparation.
- Geodatabase Automation:** Automated batch import of 200+ CSV/Excel files into a geodatabase with field mapping and data normalization; built dynamic spatial attribute updates between shapefiles.
- Geological Classification Tool:** Automated lithology and geological unit classification into 10 standard categories, ensuring data consistency and streamlined analysis.
- UTM Grid Generator:** Generated UTM coordinate grids in AutoCAD for multiple coordinate systems and projections using a custom LISP script.

Education

National Engineering Diploma - Geomatics and Topography

Ecole Supérieure de l'Aéronautique et des Technologies | Ariana, Tunisia | 2020-2023 | EQF Level 7

Language Skills

Arabic (Native) | French (Professional) | English (Professional)